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Data-driven prediction of tool wear using Bayesian regularized artificial neural networks

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ABSTRACT

The prediction of wear in cutting tools is pivotal for boosting productivity and reducing manufacturing costs. Although current data-driven models in machine learning and deep learning have advanced predictive capabilities, they often lack generality and demand substantial data training. This paper presents a novel approach using Bayesian Regularized Artificial Neural Networks (BRANNs) to precisely forecast wear in milling tools. Unlike conventional machine learning models, BRANNs merge the strengths of artificial neural networks (ANNs) and Bayesian regularization, yielding a more robust and generalized predictive model. We utilized three open-access datasets from the literature alongside an in-house dataset generated by our milling setup. Initially, we assessed the model's predictive ability by training and testing it against individual open-access datasets. We investigated the impact of input features, training data size, hidden units, training algorithms, and transfer functions on the model's predictive capability. Subsequently, we trained the model using three open-access datasets and tested it against our in-house data. Our findings demonstrate that the developed model surpasses existing state-of-the-art models in accuracy and transferability

1. Introduction

Modern manufacturing requires efficient, flexible, and cost-effective processes to ensure a competitive advantage in the market and drive continuous improvement in production. As a *milling tool* wears down, its performance and precision decline, causing a drop in product quality and a rise in production downtime and tool replacement costs. By accurately predicting tool wear, manufacturers can proactively plan tool replacements or maintenance, prevent tool failures and production interruptions, and thus maximize tools' lifespan and reduce operating expenses. Thus, the prediction of *tool wear* is crucial in optimizing production processes to ensure cost-effectiveness and efficiency.

Various models have been developed and applied to predict tool wear up to date. Typically, *Physical models* consider various factors that contribute to tool wear, such as cutting forces, temperature distribution, material properties, tool geometry, cutting speed, and feed rate [1], along with the underlying physics of the machining process, to simulate and predict the wear evolution of a cutting tool. In these models, every system component undergoes analysis to establish a physical failure model based on domain knowledge. Well-known physical models include the Taylor model [2], the Paris crack growth model [3], and the Forman crack growth model [4]. Recently, Ko and Koren [5] proposed a comprehensive physical model that relates clearance wear to the change of cutting force, making it an ideal tool for online tool wear assessment; Jawahir et al. [6] developed a new tool-life relationship that takes into account the effects of chip-groove parameters and tool coatings; Altintas et al. [7] introduced a cutting-force model that incorporates three dynamic cutting force coefficients associated with regenerative chip thickness, velocity, and acceleration terms; Qin et al. [8] presented a physics-based predictive model of cutting force in the Ultrasonic-vibration-assisted grinding (UVAG) of Ti. Still, albeit effective in predicting tool wear, these physical models have certain limitations: (1) they require expert knowledge to drive mathematical models, impeding their use by users lacking specialized expertise; (2) they often depend on predefined cutting parameters and tool geometries, delimiting their adaptability to new cutting conditions and tool designs.

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Nowadays, the development of computational power in computers and advanced software has enabled the simulation of actual machining processes under various operational conditions [9]. Specifically, the finite element (FE) simulation-based approach has been successfully applied to simulate machining processes and served as an alternative to physical experimentation, enabling cost-effective analysis. In recent years, many studies have been conducted to validate the applicability and efficiency of FE simulation in machining processes [10–16]. As a result, the FE simulation-based approach overcomes the limitations of physical models, providing accurate tool wear prediction results and a better understanding of the machining process, aiding in tool wear analysis and optimization efforts. However, the FE simulation is complex and requires expertise in computational modeling and analysis techniques. Additionally, it is computationally intensive and demands significant computational resources, such as high-performance computers and specialized software, making it too time-consuming and resource-intensive for real-time or large-scale applications.

More recently, with the advent of the fourth industrial revolution or Industry 4.0, the manufacturing industries have undergone significant changes, incorporating advanced technologies such as the Internet of Things (IoT), artificial intelligence (AI), and robotics into the manufacturing process. This integration has resulted in a new era of production, where machines and systems are interconnected, leading to more efficient, flexible, and cost-effective manufacturing [17]. A key sector that has embraced Industry 4.0 is tool condition monitoring, leading to smart monitoring systems [18] that utilize sensors and embedded systems to gather a large amount of data; such data, in turn, allow for real-time prediction of the tool condition using deep learning and machine learning.

Accordingly, significant attention has been paid to developing data-driven tool wear prediction models. Contrariwise to physical models and the FE simulation-based approach, data-driven models rely on historical data collected during machining processes rather than on closed-form mathematical equations expressing physical laws and complex, time-consuming modeling. Therefore, these models capture complex relationships in the data, even when the underlying physics is not fully understood. Moreover, data-driven models can be effective with large datasets, which allow them to learn from a wide range of tool wear scenarios. For instance, Ghosh et al. [19] developed a neural network-based sensor fusion model to estimate tool wear during the milling of C-60 steel, using the cutting force, spindle vibrations, and current signals for backward error-propagation learning. The estimation of tool wear using multiple signals exhibited good accuracy with respect to experimentally obtained values. Aghazadeh et al. [20] used a Convolutional Neural Network (CNN) to predict tool wear by collecting force and current signals; this CNN reached an accuracy of 87.2% for a system with spectral subtraction, Bayesian rigid regression, and support-vector regression. Huang et al. [21] observed that the CNN was limited in its ability to learn from high-dimensional data, and hence adopted a Deep Convolutional Neural Network (DCNN) to estimate real-time tool wear for milling, acquiring force and vibration signals to extract features in multi-domains; tool wear estimated by DCNN showed a high accuracy with an RMSE of 0.0007998 mm. Qinglong et al. [22] proposed a hybrid model, CNN-SBULSTM, to predict the remaining useful life (RUL) of milling tools; this model combines a CNN with a stacked bi-directional and uni-directional long short-term memory (SBULSTM) network; evaluated using datasets obtained from milling experiments, it could track tool wear progression and predict RUL with average prediction accuracy of up to 90%. Likewise, Cai et al. [23] presented a hybrid information model based on a long short-term memory network (LSTM) using both process information and sensor signal monitoring to predict tool wear; Experimental results on the milling data set from NASA Ames and the data set reported in the 2010 PHM Data Challenge showed the LSTM model to outperform several models, such as linear regression (LR), support vector regression (SVR), multi-layer perceptron (MLP), and CNN, in prediction accuracy. To improve CNN accuracy, Xu et al. [24] considered the significance of various features in the learning process, emphasizing the relevant ones while diminishing less useful ones during the tapping of AlSi7Mg by extracting forces and vibration signals from different channels and assigning respective weights to them; this method estimated tool wear with a maximum RMSE of 4.62. Liu et al. [25] proposed a meta-invariant feature space (MIFS) learning model to predict tool wear under cross-conditions and compared its performance to that of the DCNN and model-agnostic meta-learning (MAML); showing the feasibility and accuracy of MIFS. Xue et al. [26] proposed a Hierarchical Temporal Transformer Network (HTT-Net) that enables global modeling and long-term dependence for better tool wear state recognition by cutting signals. The findings indicated that on the PHM2010 dataset, the HTT-Net surpassed the tool wear state recognition capabilities of models from previous studies, achieving an improvement in recognition accuracy of up to 16.13% on the relevant dataset. In another work, Xue et al. [27] proposed a tool wear state recognition algorithm based on wavelet threshold de-noising (WTD), variational mode decomposition (VMD), manifold learning and K-nearest neighbor (KWRF). The results revealed that the proposed model achieved recognition accuracy of 98.74%, confirming its effectiveness. Heitz et al. [28] introduced an eXtreme Gradient Boosting algorithm for predicting the cutting forces in milling operations. The model training integrates the mechanistic force model as new features, encompassing both the time and frequency domains. The results demonstrated the impressive effectiveness of the eXtreme Gradient Boosting model, achieving an average normalized root mean square error of 14.7%, which surpasses the 21.9% obtained by the mechanistic force model. Other studies on the application of data-driven models in manufacturing can be found in [29–34]. Such data-driven models using machine learning and deep learning show their applicability in predicting tool wear; however, the models using *conventional* machine learning usually suffer from over-fitting, leading to suboptimal accuracy and generalization, while those using *deep learning* require a large amount of training data to obtain highly accurate and generalized models.

Recently, Bayesian Regularized Artificial Neural Networks (BRANN) has been applied to improve the robustness and generalization performance of neural networks, especially in scenarios with limited training data. Bayesian inference naturally incorporates regularization, which aids in preventing overfitting, making it particularly advantageous in Artificial Neural Networks (ANN). By combining the ability of ANN to learn complex patterns from the data and the ability of Bayesian regularization to address uncertainty and prevent overfitting during the training process, BRANN overcomes the shortcomings of conventional models, delivering enhanced performance predictions. Additionally, this combination is particularly beneficial in situations where datasets are limited, overcoming the limitation of deep learning models that typically require large amounts of data. Thanks to these advantages of the BRANN model, it has been successfully applied to various areas in recent years. For instance, the BRANN was applied to assess landslide susceptibility mapping in the Hoa Binh province of Vietnam [35]. A comparison of the Levenberg–Marquardt and the BRANN in this research topic demonstrated that the BRANN is more robust and efficient than the Levenberg–Marquardt network model for landslide susceptibility mapping. The BRANN was also applied to estimate the probability of default [36]. In this study, the capability of Bayesian regularization to accurately estimate network weights has been demonstrated, especially in yielding more informed and impartial credit risk decisions. Another application of the BRANN in predicting daily rainfall from sea level data has been carried out [37]. The comparative analysis of different models demonstrated that the BRANN outperformed other models such as Bayesian additive regression trees (BART), extreme gradient boosting (xgBoost), and hybrid neural fuzzy inference system (HNFIS) and extreme learning machine (ELM) in predicting the rainfall. Additionally, the BRANN was employed to predict some aspects of the gecko spatula peeling viz [38]. The results from the study showed significant potential of the BRANN model in estimating the peeling behavior. Other practical applications of BRANN can also be found in fields such as, nanotechnology [39], pile foundation engineering [40], civil engineering structures [41], ect.

The primary purpose of this work is to examine the predictive ability and generalizability of the Bayesian machine-learning method for predicting wear. Most previous studies, including those cited in this manuscript, employ the same experimental setup to both train and evaluate the predictive capabilities of their machine-learning models. In contrast, our approach involves training the model using three distinct open-access datasets and subsequently assessing its predictive performance on an independently generated in-house dataset. This methodology not only evaluates the robustness and adaptability of the model but also highlights its potential for broader application in diverse experimental conditions. From a practical standpoint, particularly for industrial applications, generating a large dataset to train a model for a specific manufacturing system is often challenging. Our methodology demonstrates the feasibility and effectiveness of using an externally sourced, open-access dataset to train the model, which is then applied to predict the outcomes of our independent, in-house experiments. Additionally, we provide an in-depth discussion on the potential causes of tool wear during the milling of Ti6Al4V, contextualizing our findings within the broader scope of machinability. We also investigate the influence of input features, training data size, hidden units, training algorithms, and transfer functions on the performance of BRANN and compare the results achieved by BRANN to those obtained from other existing models such as LR, SVR, MLP, CNN, LSTM, and MIFS, showcasing its superior accuracy and reliability.

The outline of the paper is structured as follows: Section 2 provides a theoretical basis for artificial neural networks. Following that, Section 3 describes the proposed methodology including the collection of various milling datasets used for analysis, and the proposed Bayesian regularized artificial neural networks (BRANN). Subsequently, Section 4 presents an in-depth explanation of the experimental setup, covering the dataset preprocessing and setting, as well as the implementation details. The results obtained by the proposed method using various datasets are discussed in Section 5. Lastly, significant conclusions are presented in Section 6.

2. Theoretical basis of artificial neural networks

Artificial Neural Network (ANN) is known as a computational model inspired by the structure and function of the biological neural networks in the human brain [42]. It is a type of machine learning (ML) algorithm that is capable of learning and performing tasks such as pattern recognition, classification, regression, and more [43].

The basic building block of an artificial neural network is an artificial neuron, also known as a perceptron. It takes multiple input signals, applies weights to these inputs, computes a weighted sum, and passes the result through an activation (transfer) function to produce an output. The activation function introduces non-linearity and allows the network to model complex relationships between inputs and outputs. Multiple artificial neurons are interconnected to form layers within an ANN. Typically, an ANN consists of an input layer, one or more hidden layers, and an output layer. The neurons in each layer are connected to the neurons in the subsequent layer, forming a network of interconnected nodes. A typical architecture of an ANN model is depicted in Fig. 1(a). The computing process of an ANN can be expressed by a mathematical formulation as follows [44]:

$$\mathbf{y} = f(\mathbf{W}x + b) \tag{1}$$

where $\mathbf{y}$ is the output vector; f is the activation function; $\mathbf{x}$ is the input vector; $\mathbf{W}$ and b are the weight matrix and bias parameter, respectively.

During the training phase of an ANN, the network adjusts the weights of the connections between neurons based on a specified loss (objective) function. In the context of regression problems, it is common to employ the mean square error (MSE) function as a loss function for the ANN model. The mathematical expression for MSE is typically defined as [45]:

$$L_{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(Y_i^T - Y_i^P\right)^2 \tag{2}$$

where Y_i^T represents the target corresponding to sample i, while Y_i^P refers to the output of the model for sample i. The variable N denotes the total number of samples.

The training process of an ANN is implemented by a backpropagation algorithm, which computes the gradient of the loss function with respect to the network's weights and updates them accordingly. The objective is to minimize the difference between the predicted output of the network and the desired output (i.e., minimize the MSE loss function). Once the best-trained model is obtained, an accurate output corresponding to a given input can be quickly predicted from the trained model.

Table 1
Experimental conditions of the NASA Ames milling data set.

Case	Depth of cut	Feed	Material
1	1.5	0.5	1 – cast iron
2	0.75	0.5	1 – case iron
3	0.75	0.25	1 – cast iron
4	1.5	0.25	1 – cast iron
5	1.5	0.5	2 – steel
6	1.5	0.25	2 – steel
7	0.75	0.25	2 – steel
8	0.75	0.5	2 – steel
9	1.5	0.5	1 – cast iron
10	1.5	0.25	1 – cast iron
11	0.75	0.25	1 – cast iron
12	0.75	0.5	1 – cast iron
13	0.75	0.25	2 – steel
14	0.75	0.5	2 – steel
15	1.5	0.25	2 – steel
16	1.5	0.5	2 – steel

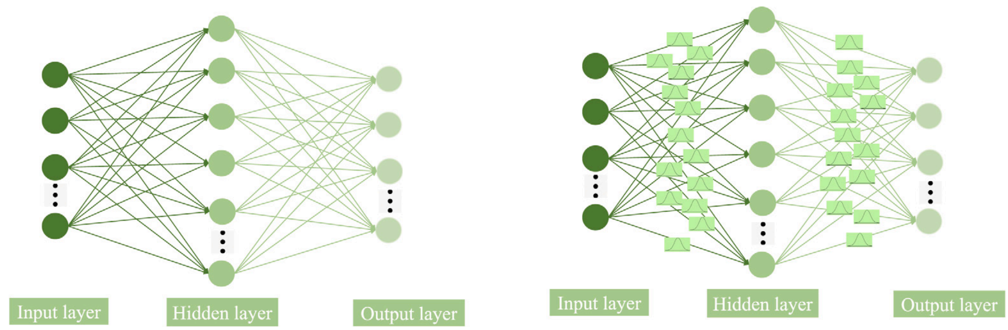
3. Proposed methodology

3.1. Data collection

In this section, four different milling datasets are introduced: the NASA Ames milling dataset (NASAAMDataset), the 2010 PHM Data Challenge dataset (PHM2010Dataset), the NUAA Ideahouse dataset (NUAADataset), and an in-house dataset from the end-milling of Ti6Al4V (InhouseDataset). Details about each dataset are provided in the following subsections.

3.1.1. NASA Ames milling data set (NASAAM Dataset)

The data were produced by performing milling of stainless steel (J45) and cast iron using 70 mm face mill cutter having 6 inserts of tungsten carbide (WC) coated with TiC/TiC-N/TiN in sequence. The experiments were performed at a cutting speed of 200 m/min, depth of cuts of 0.75 and 1.5 mm, and feeds of 0.25 mm/rev and 0.5 mm/rev. A total of 16 sets of experiments with 167 observations (samples) were carried out at different combinations of process parameters with varying number of cuts, as shown in Table 1. A vibration sensor and acoustic emission sensor were mounted to monitor the signals. The signals are then processed and used to monitor the tool wear. The flank wear (Vb) was observed and measured using a microscope for each cutting condition. Fig. 2 shows the results for Vb for all 16 cutting conditions. The data generated shows that the number of cuts was not constant for each cutting condition. Detailed information regarding the dataset can be found in [46] or at https://www.nasa.gov/content/prognostics-center-of-excellence-data-set-repository.



(a) A typical architecture of an ANN model. (b) A typical architecture of the BRANN model.

Fig. 1. A typical architecture of neural networks.

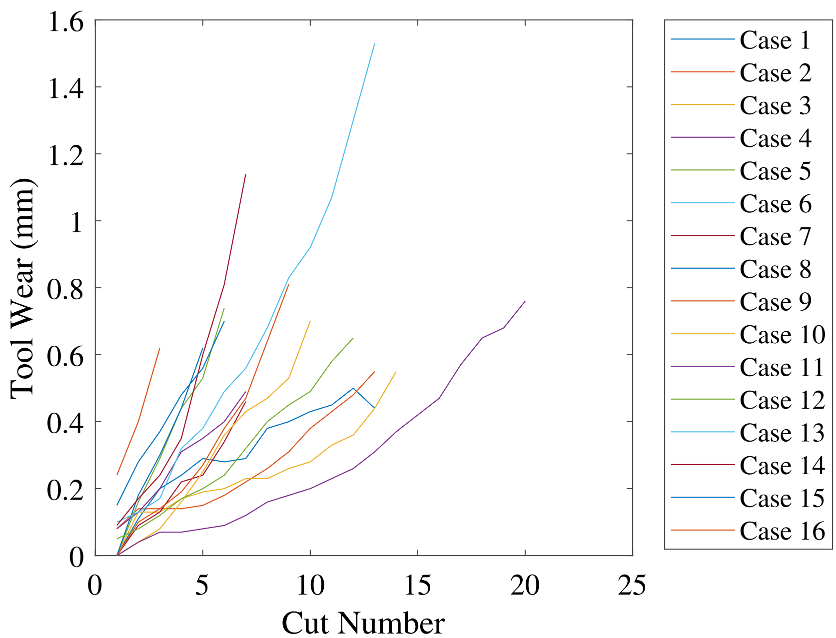


Fig. 2. Tool wear for all sixteen cases in the NASA Ames milling data set.

3.1.2. 2010 PHM data challenge data set (PHM2010 Dataset)

The data were produced by performing the milling of stainless steel (HRC 52) using a WC cutter having a three-flute ball nose. The spindle speed was 10400 rpm, the feed was 1555 mm/min, axial and radial depth of cut were 0.2 mm and 0.125 mm, respectively. A Kistler 3-component dynamometer was mounted to measure the machining forces and an accelerometer was mounted to measure the machine tool vibrations. In addition, an acoustic emission sensor was employed to monitor the high-frequency stress wave generated during the machining process. An optical microscope was used to observe the flank wear on each flute of the cutter. In this experiment, data was recorded for six individual cutters, but tool wear information is available for only three of them. These specific records, identified as C1, C4, and C6, constitute our dataset. Each record comprises 315 observations. For further details about the dataset, readers are recommended to refer to https://www.phmsociety.org/competition/phm/10.

3.1.3. Dataset of NUAA Ideahouse (NUAA Dataset)

The data set was produced to analyze the influence of process parameters on the tool wear by monitoring the signals in the milling operation [47]. The experiments were performed on titanium alloy (TC4) for the first 12 cases and on a superalloy for the thirteenth case using different tool materials and varying process parameters. The details of the experiments are presented in Table 2. Cutting force, vibration, spindle current, and power were monitored to examine the tool wear during the milling process. The monitoring signals were collected by sensors and the tool wear was observed by scanning electron microscope (SEM) after milling of each pocket layer. The maximum width of the flank wear land was chosen as per the ISO standards for tool wear criteria. In this dataset, the Vb is estimated at different spindle speeds, axial and radial depth of cuts, and workpiece materials. A total of 198 observations were recorded across 13 cases.

3.1.4. In-house dataset from end-milling of Ti6Al4V (Inhouse Dataset)

In this investigation, the end milling experiments were performed on Ti6Al4V taking TiAlN coated WC end mill cutter (Make: HB microtec GmbH, KG, Germany) having a diameter of 3 mm and 3 flutes. This study used a five-axis CNC machine tool (Haas-Multigrind CA, Trossingen, Germany) with a Siemens Sinumeric controller. The milling tests were carried out at cutting speeds of 50–95 m/min, feed per tooth of 17-50 μm/tooth, and 1 mm and 3 mm axial and radial depth of cut. The parameters were selected to perform the experiments from [48] and [49]. Fifteen experiments were conducted with a predetermined machining duration and under dry conditions. Therefore, effect of cutting fluid is not considered in this work. Out of these, four tools broke before completing the test as shown in Table 11. The Kistler 9257 dynamometer was employed to collect the force data from x, y, and z directions, as shown Fig. 3. The acoustic emission signals were collected

Table 2
The details of the cutting experiments of the NUAA Ideahouse dataset.

No.	fz (mm/r)	n (r/min)	ap (mm)	Tool diameter (mm)	Tool material
1	0.045	1750	2.5		
2	0.045	1800	3		
3	0.045	1850	3.5		
4	0.05	1750	3		
5	0.05	1800	3.5	12	Carbide Endmill without rounded corner
6	0.05	1850	2.5		
7	0.055	1750	3.5		
8	0.055	1800	2.5		
9	0.055	1850	3		
10	0.06	2200	6	12	Carbide Endmill without rounded corner
11	0.07	1300	8	20	
12	0.08	500	3	12	High speed steel
13	0.07	1200	4	12	Carbide Endmill without rounded corner

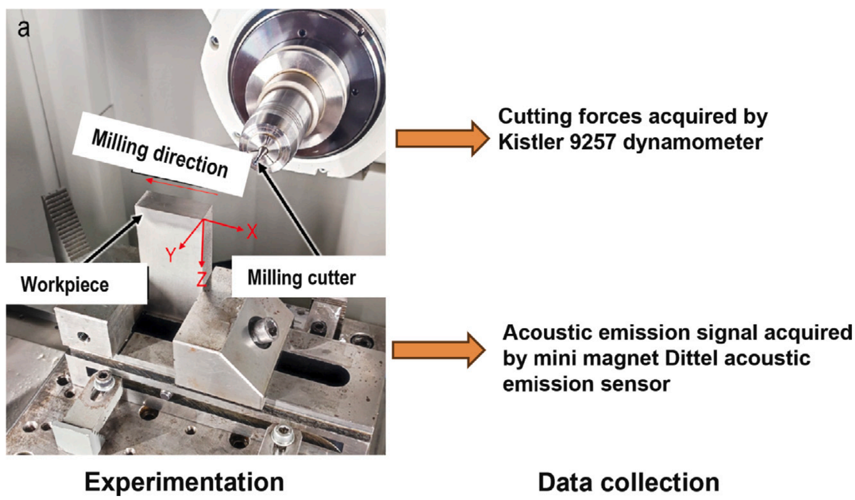


Fig. 3. Experimetal setup for inhouse performed end milling.

using the mini magnet Dittel acoustic emission sensor. It can be seen that the tool moved from the right to the left side of the workpiece. To machine a layer of 6 mm, six milling passes each with an axial depth of cut of 1 mm were performed. For each combination of cutting speed and feed, six passes were set to be performed. The signals in form of cutting forced and accoustics emissions were collected to estimate the tool wear. The sampling frequencies of the acoustic emission sensor is 1 MHz and that of the force sensor is 100 kHz. The built-up edge and tool wear were observed using VHX-5000 - Digital Microscope.

3.2. Bayesian regularized artificial neural networks

Being different from the traditional ANN, Bayesian regularized ANN (BRANN) combines the principles of Bayesian inference with ANN [50]. BRANN introduces Bayesian regularization into the training process by adding an additional term into the loss function. This additional term penalizes the presence of large weights, which is introduced to provide a smoother network response. The loss function introduced in BRANN is rewritten as follows [51]:

$$L = \beta \frac{1}{N} \sum_{i}^{N} \left(Y_i^T - Y_i^P\right)^2 + \alpha \frac{1}{N} \sum_{i}^{N} w_i^2 \tag{3}$$

where w_i is the network's weight; α and β are the hyperparameters of the loss function. If $\alpha \ll \beta$, the training algorithm will prioritize reducing errors, resulting in smaller error values. Conversely, if $\alpha \gg \beta$, training will prioritize reducing the size of weights, even if it comes at the cost of increased network errors.

In the BRANN, the weights of the neural network are considered random variables instead of fixed values. After the data is retrieved, the density function for the ANN weights can be updated according to Bayes' rule as follows [51]:

$$P(w|D,\alpha,\beta,M) = \frac{P(D|w,\beta,M)\,P(w|\alpha,M)}{P(D|\alpha,\beta,M)} \tag{4}$$

in which D is the training data set; M is a particular ANN model used; $P(D|w,\beta,M)$ denotes the likelihood function which quantifies the probability of the observed occurrence given the network weights; $P(w|\alpha,M)$ defines the prior density which characterizes our knowledge or beliefs about the weights before any data is collected; The term $P(D|\alpha,\beta,M)$ serves as a normalizing factor to ensure that the total probability sums up to 1. An illustration of the computing process of a typical BRANN model is depicted in Fig. 1(b).

By using Bayesian inference, BRANN can estimate the posterior distribution over the network's parameters rather than finding a single-point estimate. In this Bayesian framework, the optimal weights are determined by maximizing the posterior probability $P(w|D,\alpha,\beta,M)$. This maximization process is equivalent to minimizing the regularized loss function L.

The Bayesian regularization in BRANN helps to prevent overfitting by imposing a penalty on large weights, encouraging a more robust and generalized model. Additionally, the uncertainty estimates provided by BRANN can be useful in decision-making processes that require quantifying the uncertainty in predictions. BRANN has been applied in various domains, including regression, classification, and reinforcement learning tasks [52]. In this study, the BRANN is applied to predicting the milling tool wear.

4. Experimental setup

4.1. Dataset preprocessing and setting

In order to predict the tool wear using the BRANN, both process parameters (e.g., feed rate, deep of cut) and monitoring sensor signals (e.g., cutting force, torque, vibration, acoustic emission (AE), spindle power, current) are considered as inputs of the BRANN model. Since the long signals are collected from sensors for each cutting condition, a feature extraction process is performed before embedding these signals into the model with the aim of reducing the computational cost in the training process. In addition, the tool wear is measured after each cut and the tool wear at the current cut is accumulated from the preceding cut. Therefore, in order to predict the tool wear at cut n, we use the signal measured from time t_0 to time t_n instead of only using the signal measured from time t_{n-1} to time t_n for the cut n. Specifically, as described in Fig. 4, in order to predict the tool wear at cut 5, the signal recorded from t_0 to t_5 is used. For example, in the NASAAMDataset, time series sensor data consisting of 9000 time steps was recorded for each cut. Rather than merely downsampling by a factor of 2 to obtain a more suitable data length for cut n and using it to predict tool wear for that cut as in the previous study [23], we utilize the signal measured from t_0 to t_n to predict the tool wear at cut n to more accurately reflect the mechanisms operations of the machining process. Thus, to predict the tool wear at cut 1, we have the time series sensor data consisting of 9000 time steps, measured from t_0 until the completion of cut 1. For predicting the tool wear at cut 5, we use the time series sensor data consisting of 45,000 time steps, measured from time t_0 until the completion of cut 5. Similarly, this process is applied to other dataset considered in this study. Since the length of input signals for each cut varies, three statistical features, including the minimum, maximum, and mean of each input signal are extracted. Herein, we only extract the three statistical features from the sensor signals instead of extracting both complex features in frequency domain and time–frequency domain as in the previous study [23] to predict the tool wear. These extracted features are then incorporated with the process parameters to form the final input feature vector of the model. Furthermore, in order to ensure all the inputs have an equal contribution during the model's training process, the original input features are normalized by the MinMaxscaler. The mathematical expression of the MinMaxscaler is presented as follows [53]:

$$x^{norm} = \frac{x - x^{min}}{x^{max} - x^{min}} \tag{5}$$

where x^{norm} defines the normalized variable; x is the original variable; x^{min} and x^{max} are minimum and maximum variables, respectively.

For training and testing purposes, each dataset is divided into two subsets: a training set and a test set. For the NASAAMDataset, 70% of the entire dataset is used as the training set, and the remaining 30% is used for testing. For the PHM2010Dataset and NUAADataset, 80% of the dataset is used for training the model, and the remaining 20% is used for testing. When testing the model on the InhouseDataset, we use all three open-access datasets including the NASAAMDataset, the PHM2010Dataset, and the NUAADataset as the training set. Details of the different training sizes chosen for each dataset are investigated and explained in the Appendix section.

4.2. Implementation details

After preprocessing and setting the datasets, a suitable BRANN model architecture is constructed to estimate tool wear. In this study, the BRANN is implemented in MatLab software (version R2023a) using the train shallow neural network function (i.e., train function). The BRANN architecture is designed as a one-layer feed-forward network. The model has one hidden layer with thirty-two neurons, followed by a hyperbolic tangent sigmoid (tansig) transfer function. The model is trained by a Bayesian regularization backpropagation (trainbr) that updates the weight and bias values using Levenberg–Marquardt optimization. This method minimizes a combination of squared errors and weights, ultimately finding the optimal combination to ensure the network generalizes effectively. The training parameters used in the trainbr optimizer are given in Table 3. The model is trained until the stopping criterion is reached. In this study, the model is trained until the maximum training epoch reaches 1000 or the criterion of early stopping is reached. Once the best-trained model is obtained, it is saved into a *model.mat* file to reproduce the prediction results and test on the unseen data. A detailed investigation regarding number of neurons, transfer function, optimizer selection has been mentioned in Appendices A and B. The training and testing processes are conducted on a LENOVO ThinkPad T14, Gen 3, running Microsoft Windows 11 Enterprise, with an AMD Ryzen 7 PRO 6850U processor with Radeon Graphics, 2701 MHz, 8 cores, 16 logical processors, and 32 GB of RAM. The performance of the BRANN in predicting the tool wear is evaluated using three different well-known metrics including mean absolute error (MAE), root mean square error (RMSE), and the coefficient of determination (R^2). The mathematical formulation of the MAE, RMSE, and R^2 are expressed as follows:

Table 3
Setting of training parameters.

Parameter	Value
Maximum number of epochs	1000
Performance goal	0
Marquardt adjustment parameter	0.005
Decrease factor for mu	0.1
Increase factor for mu	10
Maximum value for mu	1e10
Maximum validation failures	0
Minimum performance gradient	1e−07

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|Y_i^T - Y_i^P\right| \tag{6}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(Y_i^T - Y_i^P\right)^2} \tag{7}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(Y_i^T - Y_i^P\right)^2}{\sum_{i=1}^{N}\left(Y_i^T - \bar{Y}_i\right)^2} \tag{8}$$

in which $\bar{Y}_i$ is the mean value of the targets.

Furthermore, a detailed flowchart of the application of BRANN for tool wear prediction is depicted in Fig. 5.

5. Results and discussions

In this section, the performance and applicability of the proposed BRANN model in predicting tool wear are examined by considering four distinct experimental data sets. Initially, the NASA Ames milling dataset (NASAAMDataset) including various operating conditions is considered. Subsequently, the proposed model is verified using the 2010 PHM Data Challenge dataset (PHM2010Dataset). Next, the proposed BRANN is validated by the NUAA Ideahouse tool wear dataset (NUAADataset). Lastly, an inhouse performed end milling of Ti6Al4V dataset (InhouseDataset) is taken into account to further validate the geralization of the proposed model. To demonstrate the accuracy and reliability of the proposed model, the results achieved by the proposed model are compared with those obtained from other existing models such as LR, SVR, MLP, CNN, LSTM, and MIFS. Furthermore, the impact of individual input features, training data size, hidden units, training algorithms, and transfer functions on the performance of the proposed model is also investigated.

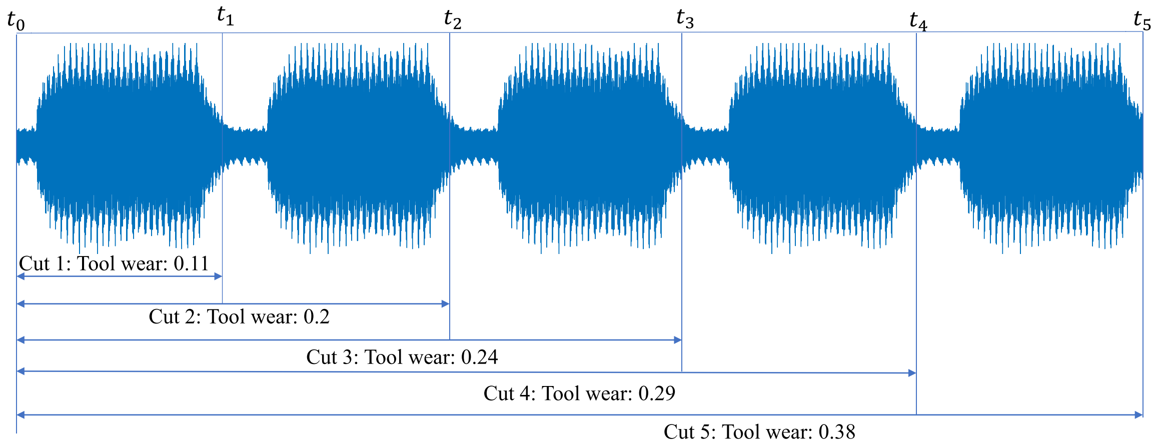


Fig. 4. Determining of a particular input signal for each cut corresponding to the tool wear.

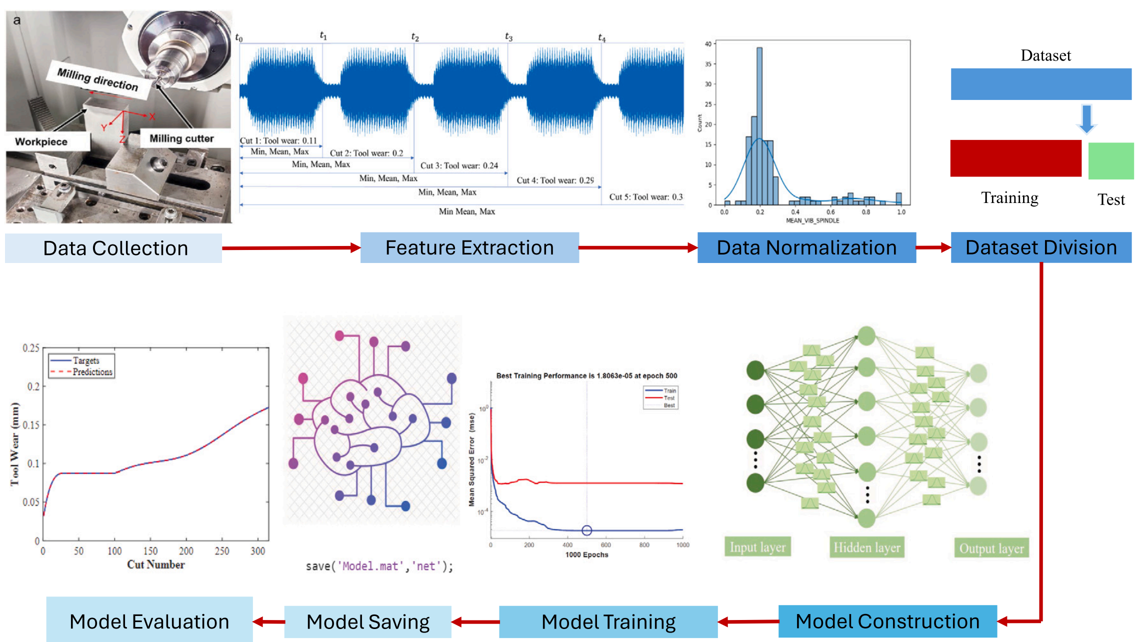


Fig. 5. Flowchart of the application of BRANN for tool wear prediction.

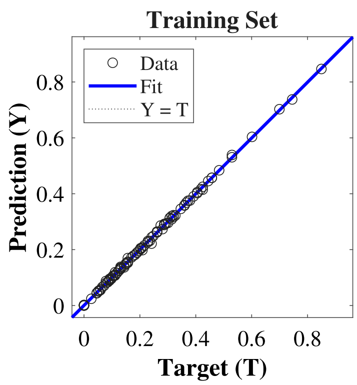
Table 4
The information of inputs and output of the BRANN model for predicting the tool wear using the NASAAMDataset.

	No.	Measured parameters	Description
Inputs	1	DOC	Depth of cut
	2	FEED	Feed rate
	3	SMCAC	AC spindle motor current
	4	SMCDC	DC spindle motor current
	5	TableVibration	Table vibration
	6	SpindleVibration	Spindle vibration
	7	AeAtTable	Acoustic emission at table
	8	AeAtSpindle	Acoustic emission at spindle
Ouputs	1	Flank wear (VB)	Flank wear, measured after runs

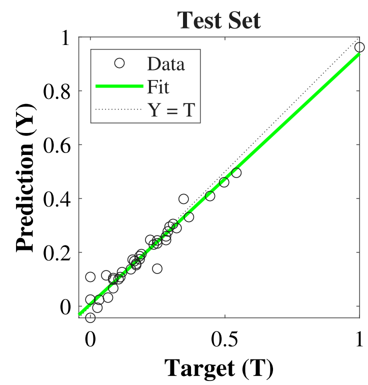
5.1. *Prediction of tool flank wear based on the NASAAMDataset*

In this case, the performance and applicability of the BRANN are validated using the NASAAMDataset. The process information and monitoring signals serve as input parameters for the model, while the output of the model is the flank wear of the tool. A comprehensive description of the input and output information of the model is found in Table 4.

To validate the accuracy and reliability of the proposed BRANN, a comparative study is conducted by comparing its results with five other existing models: LR [23], SVR [23], MLP [23], CNN [23], and LSTM [23]. The performance of these models in predicting tool wear using the NASAAMDataset is presented in Table 5. The findings demonstrate that the proposed BRANN outperforms the other five models, yielding the lowest values for MAE (0.0251) and RMSE (0.0352). Specifically, the MAE achieved by the proposed BRANN



(a) Regression results on the training set.



(b) Regression results on the test set.

Fig. 6. Regression results of the proposed BRANN model in predicting tool wear using the NASAAMDataset.

Table 5
Comparison of the performance of different models in predicting the tool wear using the NASAAMDataset.

Models	MAE	RMSE
LR [23]	0.1879	0.2146
SVR [23]	0.17	0.1945
MLP [23]	0.183	0.2168
CNN [23]	0.1835	0.2143
LSTM [23]	0.0322	0.0456
BRANN (This study)	**0.0251 (Best)**	**0.0352 (Best)**
BRANN (This study)	**0.0331 (Average)**	**0.0521 (Average)**
BRANN (This study)	**0.0065 (Variance)**	**0.0135 (Variance)**

decreases substantially compared to those of the LR (MAE=0.1879), SVR (MAE=0.17), MLP (MAE=0.183), CNN (MAE=0.1835), and LSTM (MAE=0.0322) models, with reduction factors of 7.49, 6.78, 7.29, 7.32, and 1.28, respectively. Similarly, the RMSE of the proposed BRANN also decreases significantly compared to those of the LR (RMSE=0.2146), SVR (RMSE=0.1945), MLP (RMSE=0.2168), CNN (RMSE=0.2143), and LSTM (RMSE=0.0456) models, with reduction factors of 6.1, 5.53, 6.16, 6.09, and 1.3, respectively. Moreover, it can be also seen that the average MAE and average RMSE of the proposed model after being run 10 different times are still lower than those of the LR, SVR, MLP, CNN and are competitive to those of the LSTM. Furthermore, the variances between different runs are relatively small, with a variance of MAE of 0.0065 and a variance of RMSE of 0.0135. This indicates that the prediction results obtained by the BRANN model are stable. Thus, these results show that by simply leveraging the strengths of both ANN and Bayesian regularization, the proposed approach overcomes the limitations of conventional machine learning models (e.g., LR, SVR, MLP) and deep learning models (e.g., CNN, LSTM), leading to markedly enhanced performance in tool wear prediction. This also indicates the robustness and applicability of the proposed BRANN model in accurately predicting tool wear. In addition, the regression results of the proposed BRANN model on the training and test sets are presented in Fig. 6. The depicted graphs clearly indicate that the predicted results align closely with the target values in both the training and test sets. Fig. 7 illustrates the convergence of the loss function on the training and test sets. It is seen that the model converges after 500 epochs and obtains the best training performance at epoch 500. Moreover, the tool wear prediction for different cutting conditions obtained from the proposed BRANN model is illustrated in Fig. 8. It is seen that the proposed model can predict tool wear trends and provide accurate predictions with minimal error. These results further emphasize the strong performance and practicality of the proposed BRANN model in accurately predicting tool wear.

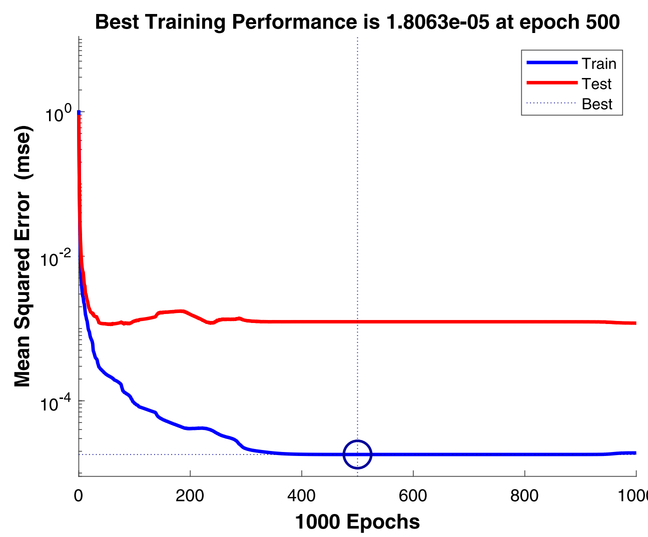


Fig. 7. The convergence of the MSE loss on training and test sets of the proposed BRANN model using the NASAAMDataset.

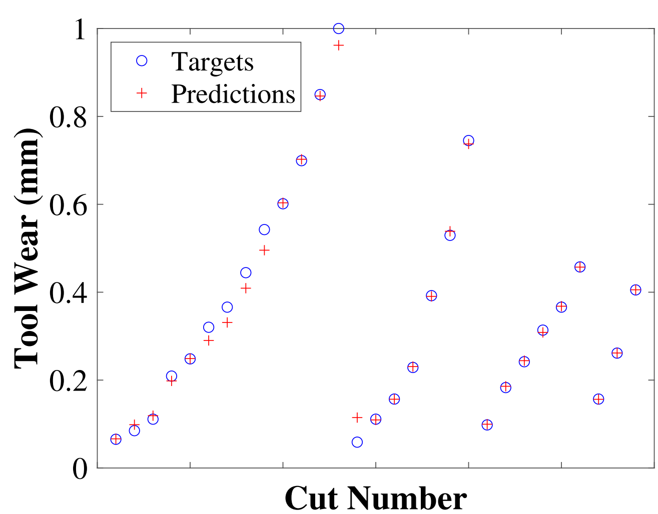


Fig. 8. Tool wear prediction obtained by the proposed BRANN model using the NASAAMDataset.

Table 6
The information of inputs and outputs for training the BRANN model for predicting tool wear using the PHM2010Dataset.

Inputs		Ouputs	
Monitoring signals	Description	Monitoring signals	Description
Force	Force (N) in X dimension Force (N) in Y dimension Force (N) in Z dimension	Flank wear	Flute-1
Vibration	Vibration (g) in X dimension Vibration (g) in Y dimension Vibration (g) in Z dimension		Plute-2
	AE-RMS (V)		Plute-3

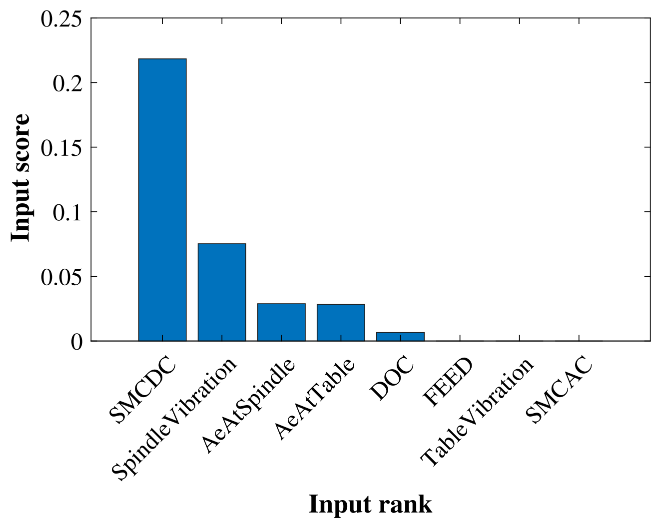


Fig. 9. Important inputs in the NASAAMDataset for predicting the tool wear.

In addition, the rank of features used as inputs of the proposed BRANN model to predict the tool wear is analyzed by a minimum redundancy maximum relevance (MRMR) algorithm [54]. This algorithm assesses relevance and redundancy using pairwise mutual information among variables and mutual information between features and the target variable. The rank score of eight inputs, which are considered as inputs for the BRANN, is depicted in Fig. 9. Notably, four inputs, namely DOC, FEED, TableVibration and SMCAC, are identified as the least significant inputs while the SMCDC, SpindleVibration, AeAtSpindle, and AeAtTable are indentified as the most significant features. Herein, the process information (i.e., DOC and FEED) is recognized as fewer impact inputs for predicting the tool wear. This is because they do not vary for each experimental condition.

Furthermore, an investigation of the effect of different hidden units, training data size, training algorithms, and transfer functions on the performance of the proposed BRANN model is given. A detailed discussion on this investigation is presented in Appendix A.

5.2. Prediction of tool flank wear based on the PHM2010Dataset

To conduct a more in-depth investigation of the performance of the proposed BRANN in predicting tool wear, the PHM2010Dataset is considered in this part. The case study involves force, vibration (in x, y, and z directions), and AE-RMS as input parameters for the model. Meanwhile, the flank wear observed at three different flutes, namely flute 1, flute 2, and flute 3 is taken as the output of the model. Table 6 provides a comprehensive overview of the input and output details to train the BRANN model.

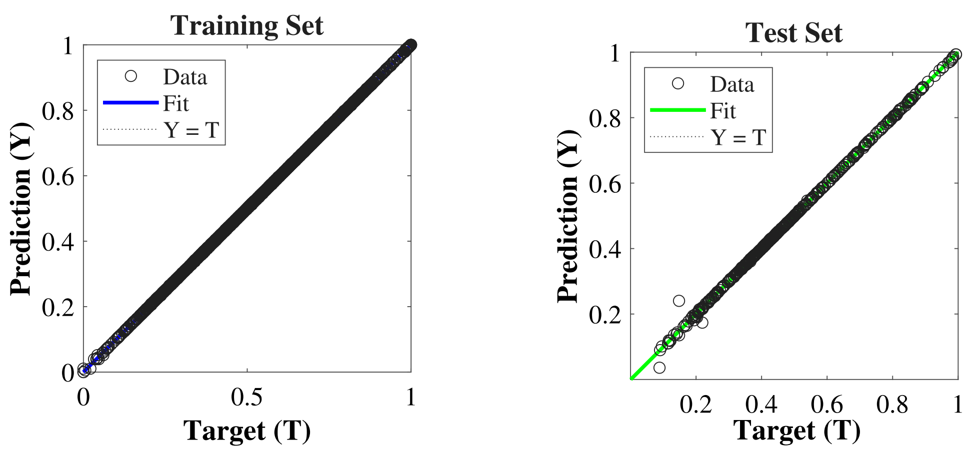
The comparative outcomes of the proposed BRANN and other existing models (e.g., LR [23], SVR [23], MLP [23], CNN [23], LSTM [23], CBLSTM [55], Deep heterogeneous GRU [56], BLSTM-PF [57], Integrated model [58] and Re-BLSTM [59]) are presented in Table 7. Evidently, the table reveals that the proposed BRANN consistently achieved the lowest MAE and RMSE values across all three scenarios (C1, C4, and C6) when compared to the other existing machine learning models (e.g., LR, SVR, MLP) and deep learning models (e.g., CNN, LSTM, CBLSTM, Deep heterogeneous GRU, BLSTM-PF, Integrated model, Re-BLSTM). In addition, it can be seen that the MAE of the proposed BRANN decreases by factors of 21.25, 1.63, 20.86 in C1, C4, and C6, respectively, compared to that of the best existing LSTM model among models compared in the literature. Similarly, the RMSE of the proposed BRANN decreases by factors of 14.25, 1.6, 2.23 in C1, C4, and C6, respectively, compared to that of the LSTM. Moreover, it can also be seen that the average MAE and RMSE of 10 training runs of the proposed BRANN model, remain lower than those of the existing models. In particular, the variance of MAE and RMSE of the 10 training runs are very small. This reaffirms the stability of the prediction results obtained by the proposed model. In addition, these results clearly demonstrate that the proposed BRANN model surpasses the other existing models in accurately predicting tool wear. These results also demonstrate that by simply leveraging the strengths of both ANN and Bayesian regularization, the proposed approach overcomes the limitations of conventional machine learning models and deep learning models, delivering significantly enhanced performance in tool wear prediction. Specifically, the proposed BRANN model helps to avoid the overfitting phenomenon of conventional machine learning during the training process and leads to a more generalized model. Moreover, BRANN is particularly beneficial in situations where datasets are limited, such as in tool wear prediction, overcoming the limitation of deep learning models that typically require large amounts of data to obtain highly accurate predictions. In addition, Fig. 10 shows the regression results of the proposed BRANN model on the training and test sets in predicting tool wear using the PHM2010Dataset. It is seen that the predicted results closely align with the experimental results (i.e., targets) in both training set and test set. Fig. 11 depicts the convergence of the loss function on both the training and test sets. It can be observed that the model converges and achieves its best training performance at epoch 1000. Furthermore, it can be seen from Fig. 12 that the proposed BRANN model can accurately predict the tool wear evolution and estimate the tool wear values with a small error in comparison with the target values obtained from the experiment. These results once again highlight the precision and applicability of the proposed BRANN model in accurately predicting tool wear.

Additionally, Fig. 13 presents an analysis of the significance of each individual inputs in the PHM2010Dataset for predicting the tool wear. It can be seen that all the features have a substantial influence on tool wear and are crucial inputs for training the BRANN model. In addition, the figure also indicates that the force signal in the z direction is the most significant factor for predicting the tool wear, whereas the vibration signal in the y direction has the least significant impact among seven considered inputs.

Furthermore, the impact of various factors such as number of hidden units, training data size, transfer functions, and training algorithms on the predictive capability of the BRANN in relation to tool wear prediction using the PHM2010Dataset is also examined. A detailed discussion is provided in Appendix B.

Table 7
Comparison of the accuracy of different models in predicting the tool wear using the PHM2010Dataset.

Models	MAE			RMSE		
	C1	C4	C6	C1	C4	C6
CBLSTM [55]	7.5	6.1	8.1	10.8	7.1	9.8
Deep heterogeneous GRU [56]	3.7	7.07	5.08	4.66	8.73	6.94
BLSTM-PF [57]	9.8	9.9	15.2	11.3	12.1	16.1
Integrated model [58]	5.54	5.27	6.86	7.67	7.84	8.6
Re-BLSTM [59]	1.707	1.963	2.630	3.100	3.930	4.807
LR [23]	0.0337	0.0164	0.0668	0.0501	0.0189	0.0904
SVR [23]	0.0098	0.0262	0.0273	0.0144	0.0293	0.0389
MLP [23]	0.0251	0.0336	0.0268	0.0288	0.0398	0.0336
CNN [23]	0.0254	0.0391	0.0383	0.0293	0.0436	0.0553
LSTM [23]	0.0085	0.0085	0.0146	0.0114	0.0117	0.0212
BRANN (This study), Best	**0.0004**	**0.0052**	**0.0007**	**0.0008**	**0.0073**	**0.0095**
BRANN (This study), Average	**0.0055**	**0.0054**	**0.0066**	**0.0123**	**0.0091**	**0.0178**
BRANN (This study), Variance	**0.0005**	**0.0005**	**0.0010**	**0.0055**	**0.0023**	**0.0083**



(a) Regression results on the training set. (b) Regression results on the test set.

Fig. 10. Regression results of the proposed BRANN model for predicting tool wear using the PHM2010Dataset.

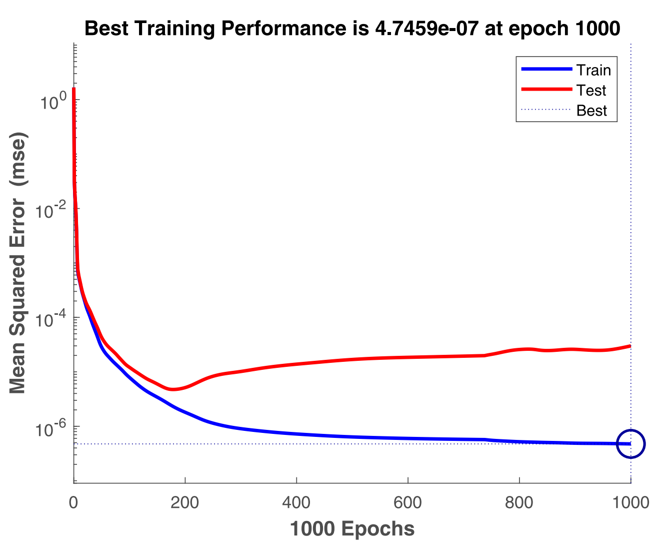
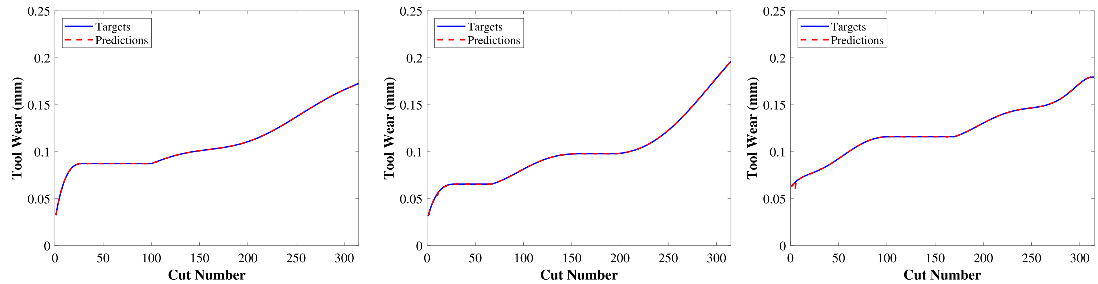


Fig. 11. The convergence of the MSE loss on training and test sets of the proposed BRANN model using the PHM2010Dataset.

5.3. Prediction of the tool wear using the NUAADataset

For further investigation the performance of the BRANN in predicting the tool wear, a dataset of NUAA Ideahouse including thirteen cutting conditions is investigated in this part. Herein, the monitoring signals for force, vibration and opc are considered as inputs of the model while the flank wear at four different edges of the tool are taken into account as the outputs of the model. The detailed description of the inputs and outputs of the model is presented in Table 8. Similar to Section 5.2, the BRANN is trained by the trainbr algorithm using 80% of the entire data. The tansig is also chosen as a transfer function of the BRANN model. The hidden units of the BRANN model is set to 16 in this case.

Table 9 illustrates a comparison between the prediction accuracy of the proposed BRANN model and the existing MIFS model [25] under various cutting conditions. The results clearly indicate that the BRANN model outperforms the MIFS model, exhibiting smaller MAE and RMSE in all cases. This signifies the superior accuracy of the BRANN model in predicting tool wear compared to the MIFS model. In addition, Table 10 presents a comparison of the RMSE of the proposed model on both training and test sets with four other existing models, including Bi-directional LSTM [60], Vanilla LSTM [60], Stack LSTM [60], and Encoder–decoder LSTM [60]. It can be observed that the proposed BRANN model achieves the lowest RMSE on the test set compared to the four aforementioned models. To be more specific, the RMSE of the BRANN model (RMSE = 0.0273) decreases significantly compared to that of the bidirectional LSTM (RMSE = 0.0401), Vanilla LSTM (RMSE = 0.0495), Stack LSTM (RMSE = 0.0491) and Encoder–decoder LSTM (RMSE=0.0362), with reduction factors of 1.47, 1.81, 1.80, and 1.32, respectively. Furthermore, the average RMSE of the proposed BRANN across 10 experimental runs remains lower than that of both Vanilla LSTM and Stack LSTM, with a small variance observed between different runs. These results again demonstrate the superior performance of the proposed BRANN model compared to other deep learning



(a) Tool wear prediction for C1. (b) Tool wear prediction for C4. (c) Tool wear prediction for C6.

Fig. 12. Tool wear predicting results obtained by the proposed BRANN model using the PHM2010Dataset.

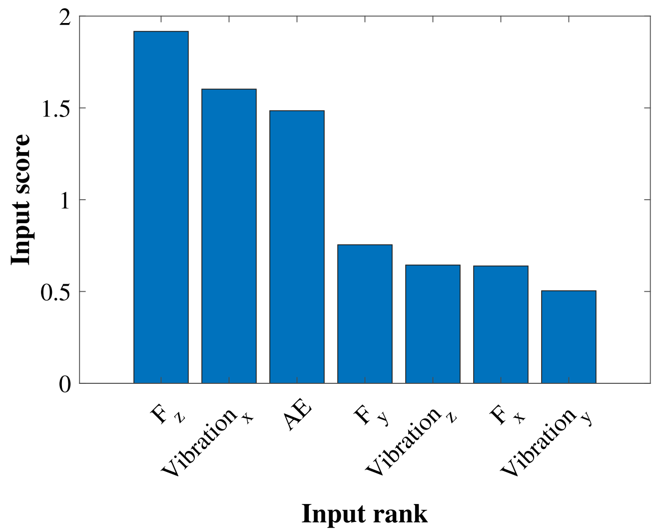


Fig. 13. Important inputs in the PHM2010Dataset for predicting the tool wear.

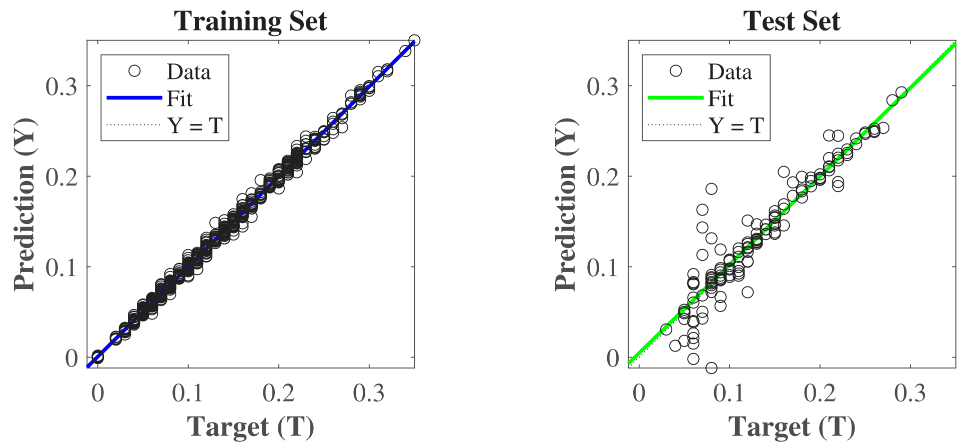
models (e.g., Bi-directional LSTM, Vanilla LSTM, Stack LSTM, Encoder–decoder LSTM). This is attributed to the fact that the proposed model does not require a large amount of data to achieve highly accurate predictions, unlike deep learning models. Consequently, these findings further validate the efficiency, stability and superiority of the proposed BRANN model over the existing approaches in tool wear prediction using the NUAADataset. Moreover, Fig. 14 shows the regression results of the proposed BRANN model on training and testing sets. The result again shows the good performance of the proposed BRANN in both training and test sets for predicting the tool wear. Fig. 15 illustrates the convergence of the loss function on both the training and test sets, showing that the model reaches convergence and achieves its best training performance at epoch 543.

Furthermore, the important inputs in the NUAADataset for predicting the tool wear are analyzed and presented in Fig. 16. It can be seen from the figure that almost inputs have certain contributions and have significant impacts on the performance of the BRANN model for predicting the tool wear. In addition, it is interesting to observe that the bending moment in y direction has the highest weight (weight = 0.46) among the eight inputs, followed by spindle power (weight = 0.29), spindle current (weight = 0.23), vibration signal at channel 2 (weight = 0.11), torsion signal in z direction (weight = 0.04), bending moment in x direction (weight = 0.03), vibration signal at channel 1 (weight = 0.0067) and axial force (weight = 0.00054).

5.4. Analysis and validation of tool condition

As discussed earlier, the end milling was performed at different combinations of process parameters taking Ti6Al4V as workpiece material. The Ti6Al4V is considered a difficult-to-cut material due to its chemical affinity and low thermal conductivity [61]. When machining Ti6Al4V, the process accelerates tool wear, leading to higher cutting forces. The experiments were conducted under dry condition, which favor the adhesion of the workpiece to the tool due to high pressure and temperature. Consequently, this adherence promotes the formation of built-up edges. As these built-up edges are expelled by the chip, they carry away some tool material, contributing to tool wear, as shown in Fig. 17. Built-up edges are accumulations of workpiece material that adhere to the cutting tool surface during machining. As the cutting process continues, these built-up edges grow in size and interfere with the cutting action, leading to poor surface finish, dimensional inaccuracies, and increased cutting forces. Moreover, when these built-up edges detach from the tool surface and are carried away by the chip flow, they drag along some portion of the tool material, contributing to progressive tool wear. Since, the milling is an intermittent cutting process, this adhesion and removal of the built-up edge enhances the chance of tool wear. In addition to that, the cyclic force acting on the cutting edge of the tool softens the tool, enabling the chance of plastic deformation [62]. Here, Systematic experiments were executed at different cutting speeds and feed rates. The tool wear is not measured but once the tool was found broken then experiment was stopped at a particular combination of process parameters. Table 11 shows the tool condition for each combination of process parameters. It is observed that for sample 1 where the cutting speed used was 50 m/min, no tool breakage was found. For samples 2, 3 and 4, where the cutting speed was 60 m/min and feed rates were 45 and 50 micron/tooth. It is noted that the tool was broken when feed rate increased. Similarly, for samples 5–11, the experiments were conducted at cutting speed of 75 m/min and feed rates varying from 22 micron/tooth to 40 micron/tooth. The tool was broken at the feed rate of 40 micro/tooth (sample 11). Likewise, the experiments at 80 m/min and 20.6–25 microns/tooth show the tool breakage for sample 14. A sudden failure of the tool is shown in Fig. 17. In all the above cases, by keeping the cutting speed constant and by increasing the feed rate, the uncut chip thickness increases, enhancing pressure acting on the tool. Moreover, the tool in the intermittent cutting action has to be prone to remove more volume of material at higher feed rate, thus the thrust force acting on the tool becomes higher. It leads to deforming the tool plastically by reducing its hardness. On the other hand, as the cutting speed increases, the corresponding feed rate at which the tool breakage is found decreases. It is due to the fact that at a higher cutting speed, the heat generated at deformation zone raises due to severe plastic deformation. The thermal conductivity of the Ti6Al4V limits the heat conduction by the chip and hence get accumulates near the cutting edge of the tool. This heat is sufficient to lower the strength of the tool and thus enhances the tool wear even at low feed rate [63].

Herein, we further validate the generalization of the BRANN model by applying it to the InhouseDataset for classifying tool conditions. Since the tool wear values are unavailable, the tool wear classification is made by training different data sets. Specifically, the BRANN model is trained on a combined dataset comprising three datasets:



(a) Regression results on the training set.

(b) Regression results on the test set.

Fig. 14. Regression results of the proposed BRANN model for predicting the tool wear using the NUAADataset.

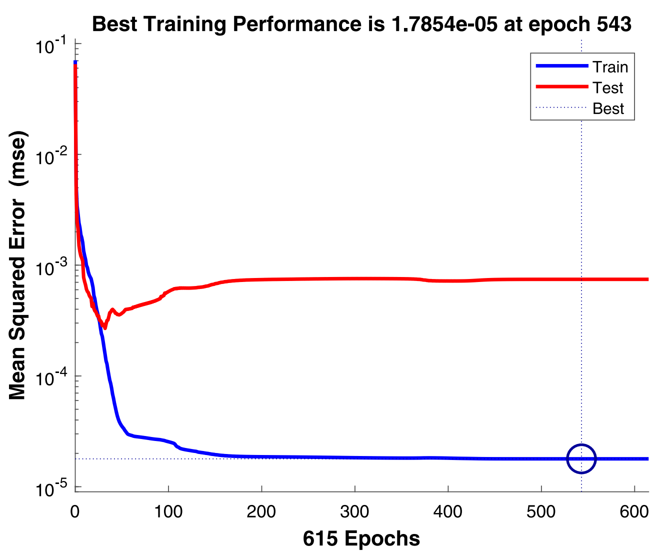


Fig. 15. The convergence of the MSE loss on training and test sets of the proposed BRANN model using the NUAADataset.

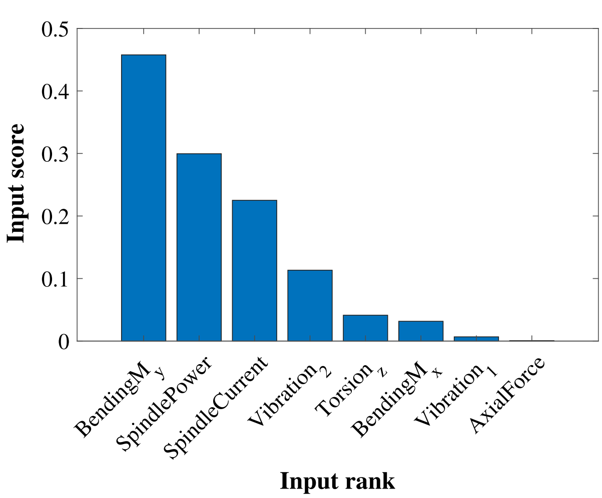


Fig. 16. Important inputs in the NUAADataset for predicting the tool wear.

the NASAAMDataset, the PHM2010Dataset, and the NUAADataset. By combining the three datasets, the input parameters of the model include all input information of the three datasets that are given in Table 4, 6 and Table 8, respectively. After training, the model is then tested on the InhouseDataset, to predict tool wear within the classification dataset. Subsequently, based on the predicted tool wear values, we classify the tool conditions into two categories: tool breakage and unbroken tool. To determine these classifications, we rely on the standard recommended value for defining a tool life criterion based on ISO 3685:1993 [64]. In the experiments, the tool wear was not measured progressively, instead, the tool breakage was considered to discard the tool. Moreover, the tool breakage happened due to excessive cutting force imposed on the cutting tool due to built-up edge formation at higher feed or cutting speed, as discussed earlier. Since the tool condition classification is done based on the tool breakage, the criteria decided is the maximum value tool flank wear. i.e., 0.6 mm. The trained BRANN model predicts the tool flank wear by using four sensor input signals, including forces in the x, y, and z directions, as well as the acoustic emission. A label tool condition classification data set containing 15 samples divided into two tool condition classes is presented in Table 11. The corresponding predicted tool flank wear results for each tool condition classification are illustrated in Fig. 18. It can be observed that the proposed model predicts tool flank wear values greater than 0.6 mm for 3 samples in the tool breakage class (except sample 11) and predicts tool flank wear values smaller than 0.6 mm for 8 out of 11 samples in the unbroken tool class. Only 3 samples (see samples 1, 6, and 9) from the unbroken tool class have predicted tool wear values greater than 0.6 mm. Thus, based on the tool life criterion, it can classify the tool conditions based on the tool wear prediction. To be more specific, as shown in Table 12, the model achieves a 75% accuracy in classifying tool breakage conditions and a 72.73% accuracy in classifying unbroken tool conditions. In general, the proposed model demonstrates a 73.33% accuracy in classifying tool conditions based on tool wear predictions, even when applied to a previously unseen data. These results demonstrate the applicability and generalization of the proposed BRANN model in evaluating the tool conditions.

Furthermore, it should be emphasized here that the BRANN model presented in this context is trained using a combination of 23 input parameters drawn from the three mentioned datasets. However, the in-house conducted end milling of Ti6Al4V dataset contains only four sensor signals (i.e., forces in the x, y, z directions, and acoustic emission) that align with the trained input parameters of the combined datasets. Consequently, when predicting tool wear results in Fig. 18, the proposed BRANN model relies only on these four inputs out of

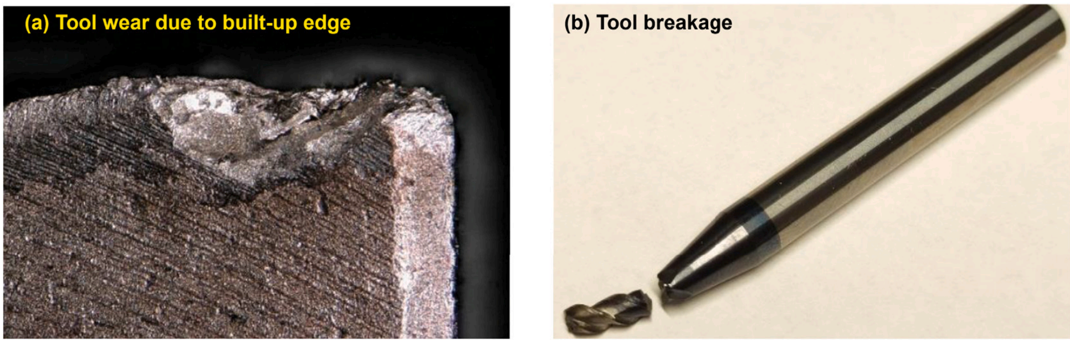


Fig. 17. Tool wear due to built-up edge and Sudden failure of tool during experiments.

Table 8
The information of inputs and outputs for training the BRANN model for predicting tool wear using the NUAADataset.

Inputs		Ouputs	
Monitoring signals	Description	Monitoring signals	Description
Force	Axial Force/N Bending Moment of X/N m Bending Moment of Y/N m Torsion of Z/N.m	Flank wear	Edge-1 Edge-2
Vibration	Channel 1 Channel 2		Edge-3
Opc	Spindle power Spindle current		Edge-4

Table 9
Comparison of the accuracy of different models in predicting tool wear using the NUAADataset under different cutting conditions.

Cutting condition	Model	MAE	RMSE
10	MIFS [25]	0.027	0.031
	BRANN	**0.0061**	**0.016**
11	MIFS [25]	0.032	0.037
	BRANN	**0.00066**	**0.00084**
12	MIFS [25]	0.063	0.027
	BRANN	**0.00080**	**0.0012**
13	MIFS [25]	0.050	0.032
	BRANN	**0.0012**	**0.0020**

Table 10
Comparison of RMSE of different models on training and test sets using the NUAADataset to predict tool wear.

Models	Training	Test
Bi-direction LSTM [60]	–	0.0401
Vanilla LSTM [60]	–	0.0495
Stack LSTM [60]	–	0.0491
Encoder–decoder LSTM [60]	–	0.0362
BRANN (This study)	**0.0042 (Best)**	**0.0273 (Best)**
BRANN (This study)	**0.0028 (Average)**	**0.0452 (Average)**
BRANN (This study)	**0.0009 (Variance)**	**0.0091 (Variance)**

the 23 available. Remarkably, despite the limited input information, the model achieves a 73.33% accuracy in classifying tool conditions. This observation highlights that the proposed model can effectively predict tool wear with commendable accuracy even when operating with restricted input data.

6. Conclusions

An effective BRANN model has been developed and successfully applied for accurately predicting tool wear. The performance and applicability of the proposed BRANN model have been demonstrated through four distinct experimental datasets including the NASAAMDataset, the PHM2010Dataset, the NUAADataset, and the dataset of inhouse performed end milling of Ti6Al4V. The outcomes of the BRANN model are compared with those of other existing models to illustrate the accuracy and reliability of the proposed model. In addition, the impact of input features, training data size, hidden units, training algorithms, and transfer functions on the performance of the BRANN model has been examined. Based on the presented theory and experimental results, some remarkable conclusions are drawn as follows:

1. The proposed BRANN model successfully avoids overfitting, leading to highly accurate tool wear predictions on both training and test sets.
2. Under the same training dataset, the proposed BRANN outperforms existing models including LR, SVR, MLP, CNN, LSTM, and MIFS in predicting tool wear in all evaluation metrics;
3. The input features, training data size, hidden units, training algorithms, and transfer functions have a significant impact on the performance of the BRANN model in predicting the tool wear;
4. The proposed BRANN model has demonstrated its applicability and generalization in predicting tool wear not only under the same cutting conditions but also under different cutting conditions.
5. By training the proposed BRANN model on a combined dataset that includes three datasets: the NASAAMDataset, the PHM2010Dataset, and the NUAADataset, the proposed model is able to achieve a 73.33% accuracy in classifying tool conditions when applied to the InhouseDataset, despite the limited input information.

Although the proposed model has demonstrated its accuracy and applicability as well as the transferability in predicting tool wear and classifying damaged and undamaged tools, there is potential for further extension. Specifically, the model could be enhanced to classify tool wear across various stages, including early wear, normal wear, and severe wear categories. Additionally, the regression prediction validation of the proposed model could be expanded to encompass diverse datasets containing tool condition variations. This study will also be expanded to include the estimation of tool wear and remaining useful tool life, taking into account process parameters and materials, as

Table 11
The tool condition for each combination of process parameter in the InhouseDataset.

Sample	Cutting speed. vc [m/min]	Feed rate fz [μm/tooth]	No of cycles	Tool breakage
1	50	30	6	NO
2	60	45	6	NO
3	60	50	6	NO
4	60	50	4	YES
5	75	22	6	NO
6	75	25	6	NO
7	75	28	6	NO
8	75	31	6	NO
9	75	34	6	NO
10	75	40	6	NO
11	75	40	6	YES
12	80	20.6	6	NO
13	80	25	6	NO
14	80	25	6	YES
15	95	17	5	YES

Table 12
Accuracy of the tool condition classification based on tool wear prediction obtained by the proposed BRANN using the InhouseDataset.

Tool condition	Classication accuray (%)
Broken tools	75
Unbroken tools	72.73
Broken and Unbroken tools	73.33

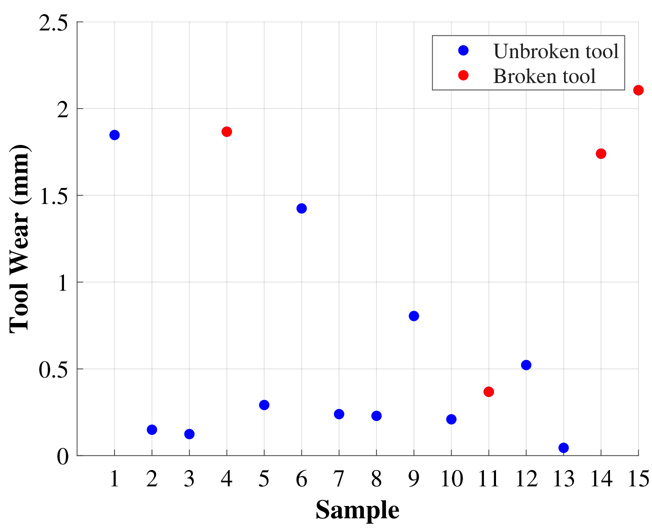


Fig. 18. Tool wear prediction obtained by the proposed BRANN using the InhouseDataset.

future work. Moreover, regarding technical limitations, the proposed approach is still limited in estimating tool wear where data are very limited, detecting a single component involving multiple faults, and estimating in environments with few data. Addressing these aspects will open up new enhancements for the current study in the future.

CRediT authorship contribution statement

Tam T. Truong: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Jay Airao:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Bahman Azarhoushang:** Writing – review & editing, Investigation, Data curation. **Panagiotis Karras:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Ramin Aghababaei:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ramin Aghababaei reports financial support was provided by Villum Foundation. Ramin Aghababaei reports a relationship with Villum Foundation that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.measurement.2024.115303.

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